Lab 7 — Fuzzing for Input Validation Bugs

Software Engineering · Traffic Fine input processor

STUDENT	SRN	PROJECT	RUNTIME
Dhruv Swatantramath	PES1UG24CS155	Traffic Fine	Python 3.13.15

Tested functions and properties

FUNCTION	HYPOTHESIS COVERAGE

`parse_ticket_id`	`test_parse_ticket_id_returns_normalized_components`
	`test_parse_ticket_id_rejects_zero_year`
	`test_parse_ticket_id_rejects_unicode_and_wrong_separators`
	`test_all_functions_reject_none`

`total_fines`	`test_total_fines_sums_nonnegative_amounts_and_caps`
	`test_total_fines_treats_empty_and_whitespace_as_zero`
	`test_total_fines_rejects_negative_amounts`
	`test_total_fines_rejects_unicode_or_malformed_tokens`
	`test_all_functions_reject_none`

`speed_excess`	`test_speed_excess_is_nonnegative_and_matches_integer_percentage`
	`test_speed_excess_rejects_zero_limit`
	`test_speed_excess_rejects_negative_speed`
	`test_speed_excess_rejects_malformed_unicode_readings`
	`test_all_functions_reject_none`

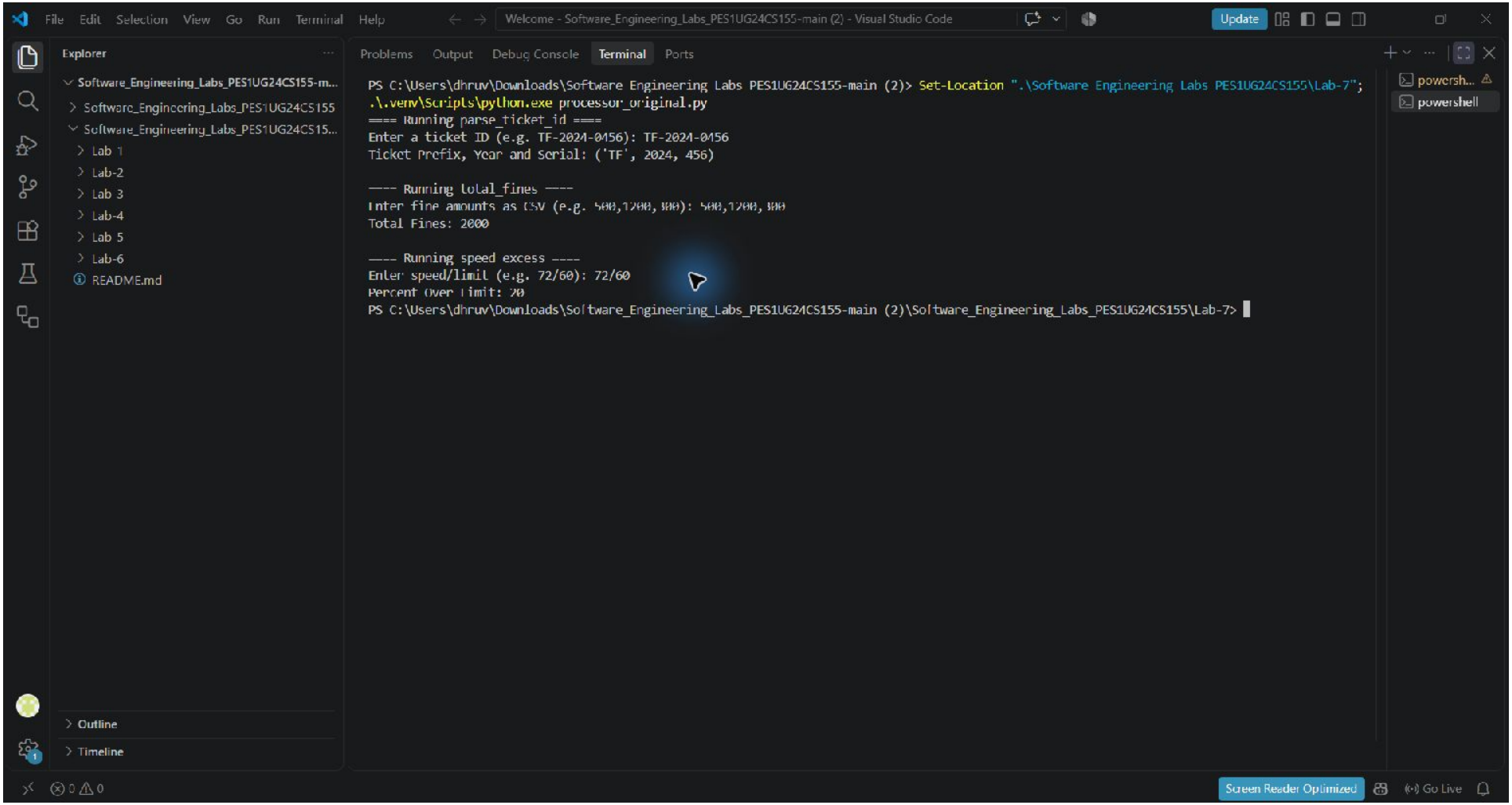
Workflow and verified results

The unmodified starter was kept as `processor_original.py`. The same Hypothesis suite selects it with `LAB7_USE_ORIGINAL=1` for the baseline, then runs against the corrected `processor.py`. The baseline collected 14 test cases (10 failed, 4 passed). The final run collected 14 and passed all 14. The examples use Python 3.13.15, pytest 9.2.0, and Hypothesis 6.170.15.

The fix validates input types and formats, rejects a zero ticket year and negative fines, treats blank fine input as zero, requires a positive speed limit, and returns zero when speed is at or below the limit.

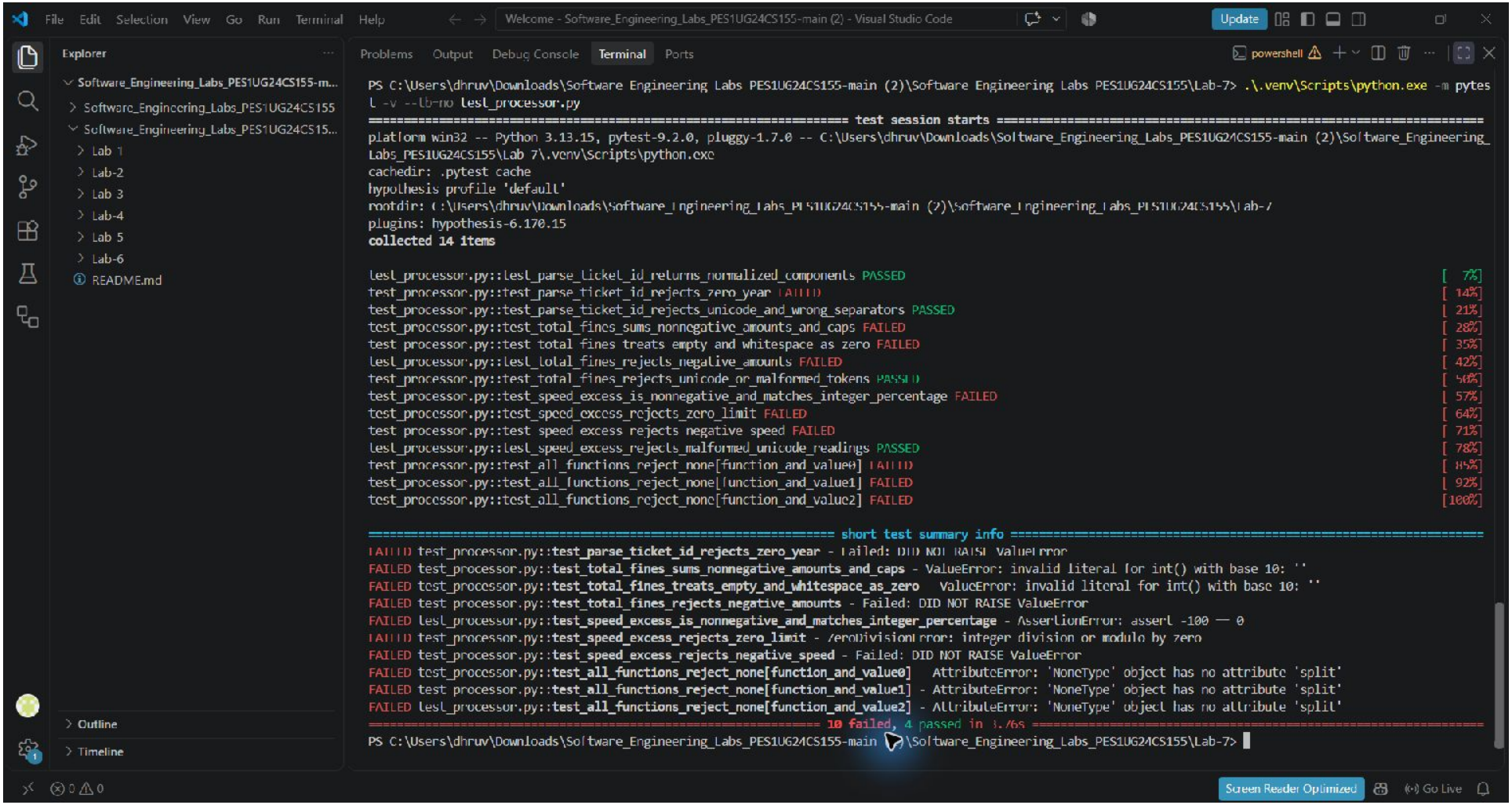
1. Original application

The unmodified starter is run with ticket ID `TF-2024-0456`, fine values `500,1200,300`, and reading `72/60`. It prints the parsed ticket, 2,000 total fines, and 20 percent over the limit.



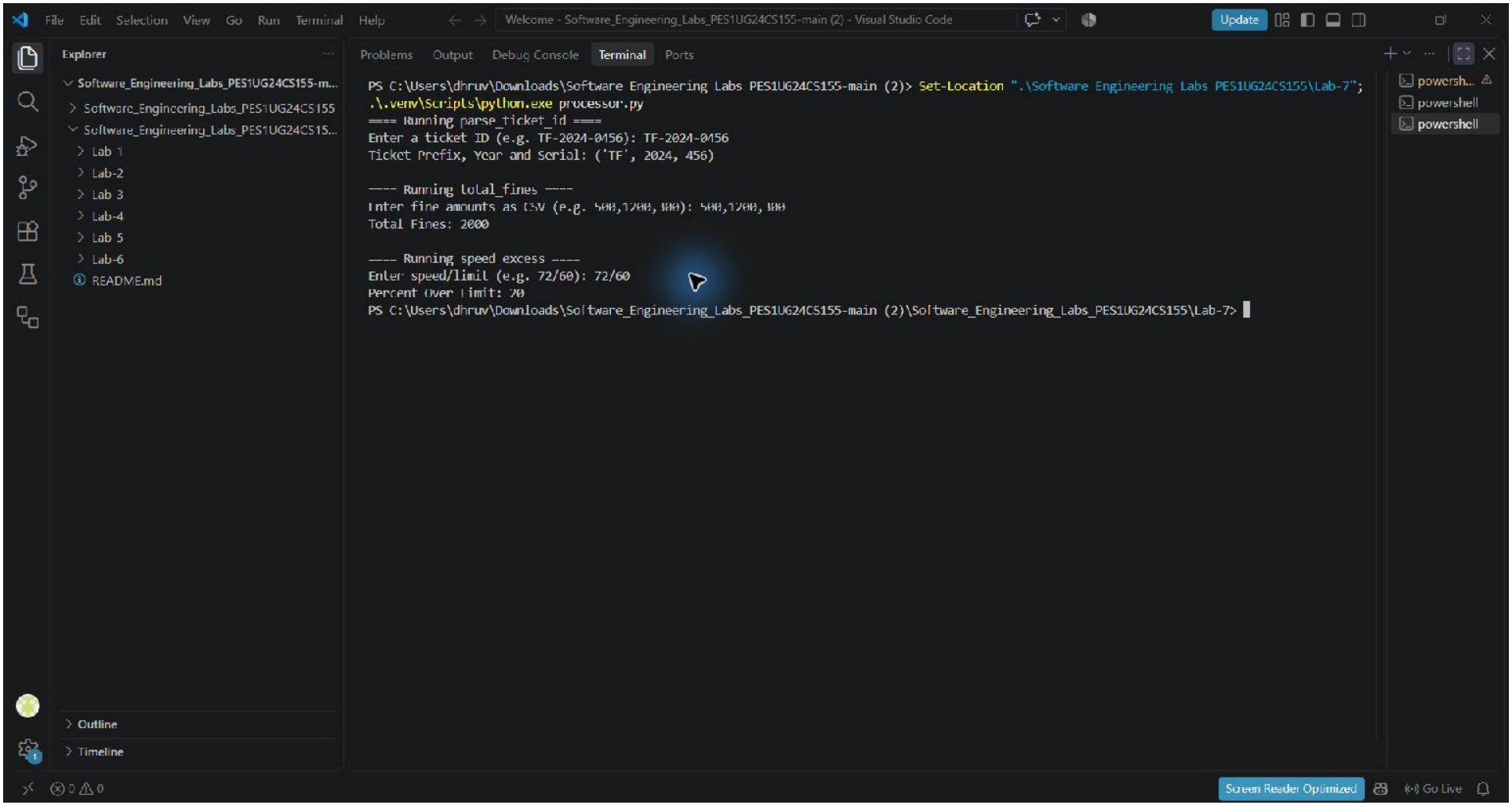
2. Initial Hypothesis run against the original

The original module was selected with `LAB7_USE_ORIGINAL=1`. Pytest collected 14 cases: 10 failed and 4 passed. Failures expose zero-year acceptance, negative-fine acceptance, incorrect at-or-under-limit behavior, zero-limit division, and missing type validation.



3. Fixed application

The corrected `processor.py` is run with the same sample inputs. Its output keeps the valid ticket and fine total while reporting 20 percent over the limit.



4. Final Hypothesis run against the fixed implementation

The original-module environment switch was cleared. All 14 collected pytest cases passed against `processor.py`.

